第2讲 基本框架

2.1 (a) 在状态a时，证券1和2分别支付1和1；b时为1和2；c时为1和3。

(b) $X \equiv \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$ $(\Omega \times N)$

(c) $\theta = [\theta_1, \theta_2]^T$, $x = X\theta = [\theta_1+\theta_2, \theta_1+2\theta_2, \theta_1+3\theta_2]^T$

$S_1\theta_1 + S_2\theta_2$

(d) $\theta_M = \sum_{k=1}^{K} \bar{\theta}_k$，有 $\bar{\theta}_k = [1, 2]^T$，有 $\theta_M = [K, 2K]^T$

$X\theta_M = [K+2K, K+4K, K+6K]^T = [3K, 5K, 7K]^T$

$V = S_1K + 2S_2K$

(e) $M = \{x : x = X\theta : \theta \in R^2\} = \{[\theta_1+\theta_2, \theta_1+2\theta_2, \theta_1+3\theta_2]^T : \theta_1, \theta_2 \in R^2\}$

(f) $M = \{[\theta_1+\theta_2, \theta_1+2\theta_2, \theta_1+3\theta_2 : \theta_1, \theta_2 \geq 0\} / \theta \in R_+^2$

(g) $X = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 2 & 0 \\ 1 & 3 & 1 \end{bmatrix}$ $r(X) = 3$, $\Rightarrow M = R^3$

2.2 $B(e, \{X, S\}) = \{C \geq 0 : C_0 = e_0 - S^T\theta, C_1 = e_1 + X\theta, \theta \in R^2\}$

$\bar{\theta} = [\theta_1, \theta_2]^T$，有 $e_0 = S^T\bar{\theta}$，有 $C_0 = e_0 - S^T\theta = S^T(\bar{\theta} - \theta)$ $\theta \in R^2$

$B(\theta) = \{S \cdot \theta' \leq S \cdot \bar{\theta} = S_1\theta_1 + S_2\theta, \theta' \in R^2\}$

2.3 $B(\theta) = \{S \cdot \theta' = S_1\theta_1' + S_2\theta_2' + S_3\theta_3' \leq S \cdot \bar{\theta} = S_1\theta_1 + S_2\theta_2, \theta' \in R^3\}$

当 $\theta_1' = \theta_1$, $\theta_2' = \theta_2$, $\theta_3 = 0$ 时，不等式成立。

2.4 (a) $c = [100; 1]$, $U^a = 2\log 10$

(b) $B(e, \{X, S\}) = \{C \geq 0 : C_0 = e_0 - S^T\theta, C_1 = e_1 + X\theta, \theta \in R^1\}$

$C_0 = 100 - \theta$, $C_1 = 1 + (1+r_F)\theta$, $W = 100 - \theta + \frac{1+(1+r_F)\theta}{1+r_F} = 100 + \frac{1}{1+r_F}$

b2. $U(C_0, C_1) = \log(100-\theta) + \rho\log[1+(1+r_F)\theta]$ 令 U_{max} 有 U^b.

$S = e_0 - C_0 = \theta \Rightarrow C_0/S = \frac{100}{\theta} - 1$, $C_1 = 1 + (1+r_F)(100 - C_0)$

有 $\theta=\frac{100\rho(1+r_F)-1}{(1+\rho)(1+r_F)}$, $C_0=\frac{w}{1+\rho}$ $\Rightarrow$ $C_1=\frac{\rho(1+r_F)w}{1+\rho}$

有 $U^b(C_0,C_1)=\log\frac{w}{1+\rho}+\rho\log\frac{\rho(1+r_F)w}{1+\rho}$

b3. $\theta=\frac{100\rho}{1+\rho}-\frac{1}{(1+\rho)(1+r_F)}$，与 r_F 成正比，与 ρ 成正比

(c) $w>100$，令 $\theta=0$，$U^c=U^a\le U^b$

(d) $U^b(w-g)\ge U^a$ 令等号成立时 g 最大

有 $g=w-(1+\rho)\cdot 100^{\frac{1}{1+\rho}}[\rho(1+r_F)]^{\frac{\rho}{1+\rho}}+100+\frac{1}{1+r_F}$

有 $\frac{\partial g}{\partial \rho}=\frac{100}{1+\rho}[100\rho(1+r_F)]^{-\frac{\rho}{1+\rho}}\ln[100\rho(1+r_F)]$

$\frac{\partial g}{\partial r_F}=\frac{[100\rho(1+r_F)]^{\frac{1}{1+\rho}}-1}{(1+r_F)^2}$

有
$$\begin{cases}\rho(1+r_F)>\frac{1}{100}\Rightarrow \frac{\partial g}{\partial \rho}>0,\ \frac{\partial g}{\partial r_F}>0\\ \rho(1+r_F)=\frac{1}{100}\Rightarrow \frac{\partial g}{\partial \rho}=0,\ \frac{\partial g}{\partial r_F}=0\\ \rho(1+r_F)<\frac{1}{100}\Rightarrow \frac{\partial g}{\partial \rho}<0,\ \frac{\partial g}{\partial r_F}<0\end{cases}$$